

Researchers: Giant Sloths, Mastodons Lived with Humans in Americas

A sloth is a furry animal that lives mainly in South and Central America. It moves very slowly and spends most of its time in trees.

But thousands of years ago their ancestors were huge. Giant sloths could weigh up to 3.6 metric tons and lived on the ground.

For many years, scientists believed the first humans to arrive in the Americas quickly killed off these giant ground sloths, along with many other huge animals. Those include mastodons, saber-toothed cats, and dire wolves.

But research in recent years suggests that humans might have arrived in the Americas thousands of years earlier than scientists had once believed. New findings suggest that humans lived with the big animals for thousands of years.

This illustration shows large animals which once roamed prehistoric North and South America. Top row from left, a glyptodon, a lestodon, and a horse. Bottom row from left, a mastodon, a saber-toothed cat and a toxodon. (Mauro Muyano via AP)

Daniel Odess is an archaeologist at White Sands National Park in the American state of New Mexico.

“There was this idea that humans arrived and killed everything off very quickly — what’s called ‘Pleistocene overkill,’” he said. But new discoveries suggest that “humans were existing alongside these animals for at least 10,000 years, without making them go extinct.”

Santa Elina is a place in central Brazil where archeologists are looking for the remains of ancient animals and humans. There, scientists have found bones of giant sloths. However, the bones look like humans used them and changed them.

Mírian Pacheco is a researcher in a laboratory at the University of São Paulo. Pacheco recently showed The Associated Press a small, round sloth fossil. She noted that the fossil is smooth and there is a very small hole near one edge. She said it looks like humans changed the bone on purpose. She added that researchers think the bone was probably used as jewelry.

Researcher Mírian Pacheco holds a sloth fossil dated to around 27,000 years ago, at the University of São Paulo, Brazil, on Sept 2, 2024. (AP Photo/Christina Larson)

The animal bones from Santa Elina are about 27,000 years old. That is older than scientists had thought possible. Some had believed that humans only arrived in the Americas 11,000 years ago.

Researchers at first wondered if humans had been working with ancient fossils. But Pacheco's research strongly suggests that ancient people were carving "fresh bones" shortly after the animals died.

Pacheco studied chemical changes that take place when a bone becomes a fossil. She said the bone had been carved "before the fossilization process." Pacheco's team also ruled out natural processes.

In high school, Pacheco learned that most archeologists believed humans arrived in the Americas about 13,000 to 11,000 years ago. "What I learned in school was that Clovis was first," she said.

Clovis is a place in New Mexico where archaeologists in the 1920s and 1930s found objects dated to between 11,000 and 13,000 years ago. Until more recent years, Clovis objects, or artifacts, were among the oldest known in the Americas.

Scientists say there was a large decrease in the number of large animals at that time. The period is also believed to mark the end of the last Ice Age: a period when large areas of land were covered with thick ice and snow.

Paleontologist Thaís Pansani stands in front the reconstructed skeleton of a giant ground sloth at the Smithsonian National Museum of Natural History in Washington, on July 11, 2024. (AP Photo/Mary Conlon)

Briana Pobiner is a paleontologist with the Smithsonian Institution's Human Origins Program. Pobiner said, "It was a nice story for a while, when all the timing lined up," adding, "But it doesn't really work so well anymore."

In the past 30 years, new research methods, including ancient DNA, have suggested a different story.

The first place that most scientists agreed was older than Clovis was Monte Verde, Chile.

There, researchers discovered 14,500-year-old stone tools, pieces of animal skins, and several plants people could eat and use for medicine.

"Monte Verde was a shock," said Tom Dillehay. He is an archeologist at Vanderbilt University in the American state of Tennessee. Dillehay has carried out research at Monte Verde for many years.

This undated photo made available by the National Park Service in September 2021 shows fossilized human fossilized footprints at the White Sands National Park in New Mexico. (NPS via

At New Mexico's White Sands, researchers have uncovered human footprints dated to between 21,000 and 23,000 years ago. The site also has tracks of giant animals dated to about the same time. However, archeologists have not found any artifacts at the site.

Researchers continue to disagree about the timing of humans' first arrival in the Americas. But if they did arrive earlier than once thought, several scientists now believe that it is possible they did not immediately kill off the giant animals living in the Americas.

I'm Andrew Smith. And I'm Mario Ritter, Jr.

Christina Larson reported this story for The Associated Press. Andrew Smith adapted the story for Learning English.

Vocabulary

furry - adj. ที่ปกคลุมด้วยขนสัตว์

Pleistocene - n. ยุคหนึ่งของเวลาตั้งแต่ประมาณ 2.5 ล้านปีถึง 11,000 ปีที่แล้ว

extinct - adj. ไม่มีอยู่บนโลกอีกต่อไป

archeologist - n. ผู้ที่ศึกษาชีวิตมนุษย์โบราณ

fossil - n. ซากของสิ่งมีชีวิตที่กลายเป็นหิน

jewelry - n. วัตถุที่สวมใส่บนร่างกายเพื่อความสวยงามและแสดงสถานะ

carve - v. การค่อยๆ ตัดวัสดุออกเพื่อสร้างลวดลายหรือพื้นผิวพิเศษ

Reading Comprehension Quiz

1. According to the article, what is a primary characteristic of a modern sloth?

- A) It is a reptile.
- B) It is a fast-moving ground animal.
- C) It is a furry animal that lives mostly in trees and moves slowly.
- D) It is an ancient creature that is now extinct.

2. How did giant sloths differ significantly from modern sloths?

- A) They lived only in North America.
- B) They were much smaller and faster.
- C) They were huge, weighing up to 3.6 metric tons, and lived on the ground.
- D) They were known for their colorful fur.

3. What was the long-held belief among scientists regarding the first humans in the Americas and giant ground sloths?

- A) Humans coexisted peacefully with giant sloths for thousands of years.
- B) Humans introduced giant sloths to the Americas.
- C) Humans quickly caused the extinction of giant ground sloths and other large animals.
- D) Giant sloths died out naturally due to climate change.

4. What do recent research findings suggest about the timing of human arrival in the Americas?

- A) Humans arrived much later than previously thought.
- B) Humans arrived thousands of years earlier and lived alongside large animals for a long time.
- C) Humans only arrived after all giant animals had become extinct.
- D) Humans never actually encountered giant animals.

5. What does the term 'Pleistocene overkill' refer to, according to Daniel Odess?

- A) A theory that giant animals overpopulated the Earth during the Pleistocene era.
- B) The idea that a massive meteor strike caused widespread extinctions.
- C) The belief that humans arrived and rapidly caused the extinction of many animals.
- D) A natural phenomenon where animal populations died off due to harsh weather.

6. What significant discovery was made at Santa Elina in central Brazil?

- A) Undisturbed giant sloth skeletons.
- B) Bones of giant sloths that appear to have been used and altered by humans.
- C) Early human settlements without any animal remains.
- D) Evidence of giant sloths cultivating plants.

7. What did Mírian Pacheco suggest about the small, round sloth fossil she examined?

- A) It was a naturally occurring rock formation.
- B) It showed no signs of human intervention.
- C) It was intentionally changed by humans, possibly for use as jewelry.
- D) It was part of a sloth's digestive system.

8. Why is the age of the animal bones from Santa Elina (around 27,000 years old) particularly noteworthy?

- A) It suggests giant sloths lived longer than previously thought.
- B) It confirms the old belief that humans arrived only 11,000 years ago.

- C) It is much older than the previously accepted timeline for human arrival in the Americas.
- D) It indicates that Santa Elina was a major burial site for animals.

9. What did Pacheco's research strongly suggest about how ancient people worked with bones?

- A) They only collected already fossilized bones.
- B) They primarily used bones for building shelters.
- C) They carved 'fresh bones' shortly after the animals died.
- D) They had no interest in animal bones.

10. How did Pacheco's team determine that the bone from Santa Elina was carved before fossilization?

- A) By observing the type of tools used for carving.
- B) By studying chemical changes that occur during fossilization and ruling out natural processes.
- C) By comparing its size to other known artifacts.
- D) By carbon-dating the specific marks on the bone.

11. What was Mírian Pacheco's initial understanding about human arrival in the Americas, based on her high school education?

- A) Humans arrived about 27,000 years ago.
- B) Clovis was the first evidence of human presence, dating back 13,000 to 11,000 years ago.
- C) Humans coexisted with giant sloths.
- D) There was no consensus on the timing of human arrival.

12. What is significant about the Clovis site in New Mexico?

- A) It is where the oldest known giant sloth fossils were found.
- B) It provided the first evidence of human footprints in the Americas.
- C) It yielded objects, or artifacts, dated to between 11,000 and 13,000 years ago, which were among the oldest known.
- D) It is where scientists first discovered evidence of 'Pleistocene overkill.'

13. What two major events are believed to have occurred around the same time as the Clovis period (11,000-13,000 years ago)?

- A) The arrival of modern sloths and the beginning of agriculture.
- B) A large decrease in the number of large animals and the end of the last Ice Age.
- C) The discovery of fire and the invention of the wheel.
- D) The first human settlements and the formation of continents.

14. What is Briana Pobiner's current perspective on the 'nice story' (the old belief about human arrival and its impact)?

- A) She believes it is still the most accurate explanation.
- B) She thinks it works perfectly well with all new evidence.
- C) She suggests it no longer aligns well with new discoveries.
- D) She has no strong opinion on the matter.

15. Which site was the first that most scientists agreed was older than Clovis?

- A) White Sands National Park.
- B) Santa Elina.
- C) Monte Verde, Chile.
- D) The Smithsonian National Museum.